Project - Low Level Design

on

Entertainment Content Generator

Course Name: Gen AI

Institution Name: Medicaps University – Datagami Skill Based Course

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1. Introduction

The **Entertainment Content Generator** is a Generative AI-based system designed to assist entertainment professionals in generating screenplay scenes automatically from simple textual inputs. The system utilizes modern AI technologies including Large Language Models (LLMs), prompt engineering, and a vector database to generate structured and high-quality script content.

The system follows a **Retrieval-Augmented Generation (RAG)** architecture in which relevant domain knowledge is retrieved from a vector database before sending the prompt to the LLM. This ensures that the generated output maintains professional screenplay formatting and industry-standard storytelling techniques.

The application consists of a **React-based frontend**, a **FastAPI backend**, a **vector search module using FAISS**, and **LLM integration using the Google Gemini API**.

1.1 Scope of the Document

This document describes the **Low Level Design (LLD)** of the Entertainment Content Generator system.

The scope includes:

- System architecture
- Module design
- API design
- Component interactions
- Data flow within the system
- UML diagrams describing the system structure and behavior

1.2 Intended Audience

This document is intended for:

- Software developers
- System architects
- Project evaluators
- Academic supervisors

1.3 System Overview

The Entertainment Content Generator allows users to generate screenplay scenes based on user input parameters such as topic, genre, mood, and scene length.

The system performs the following operations:

1. User provides input through the frontend interface.
2. The frontend sends the request to the backend API.
3. The backend retrieves relevant screenplay guidelines using FAISS.
4. The prompt engine constructs a structured prompt.
5. The prompt is sent to the LLM (Gemini API).
6. The LLM generates a screenplay scene.
7. The generated scene is returned to the frontend for display.

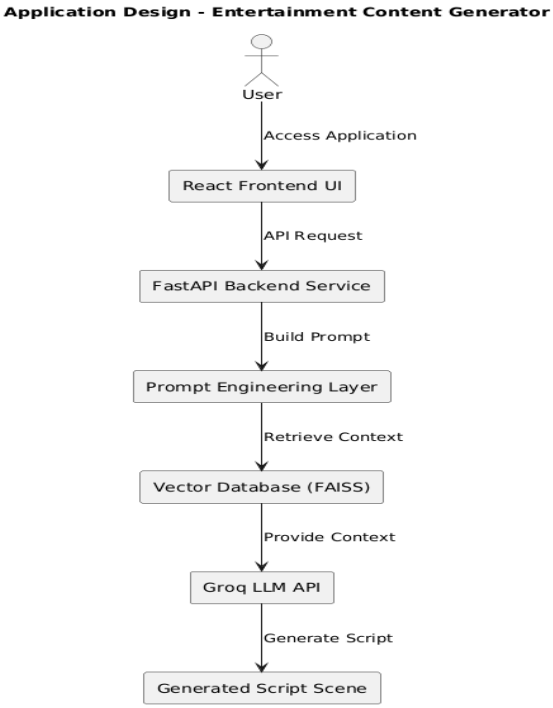
2. System Design

The system is designed using a **modular architecture** to improve scalability, maintainability, and extensibility.

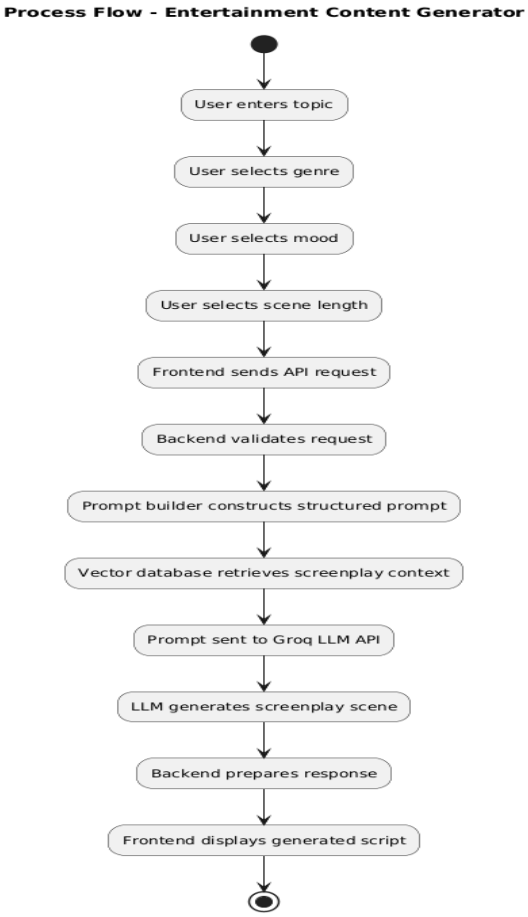
Major layers include:

- Presentation Layer
- Application Layer
- Retrieval Layer
- AI Generation Layer
- Data Layer

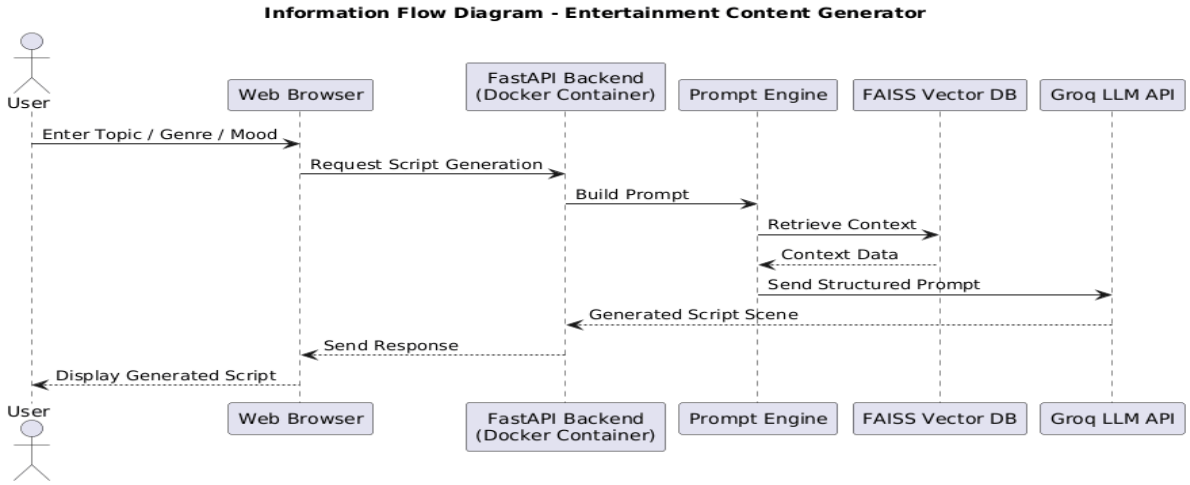
2.1 Application Design



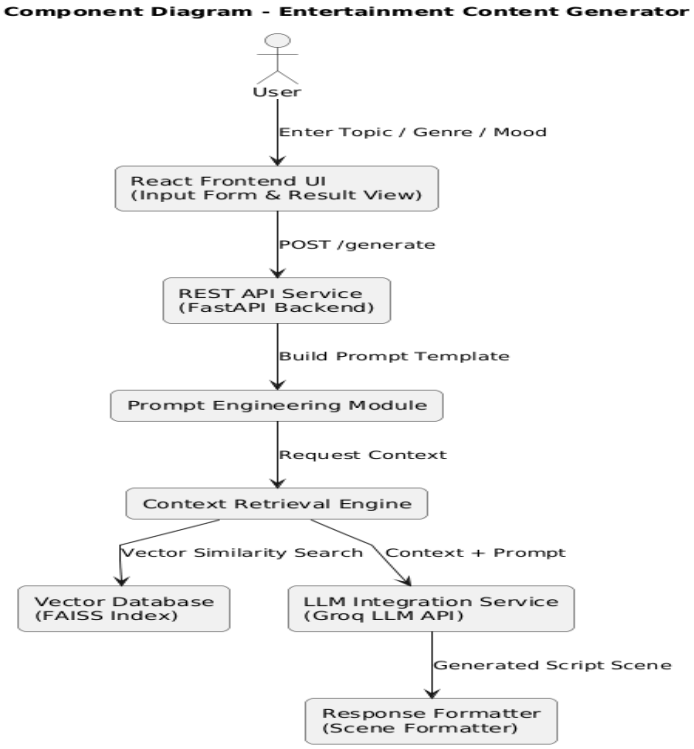
2.2 Process Flow



2.3 Information Flow



2.4 Components Design



2.5 Key Design Considerations

Scalability : The modular architecture allows each component to scale independently.

Maintainability : The system separates responsibilities across multiple modules, making maintenance easier.

Performance : The use of FAISS enables fast similarity search, while Groq provides high-speed inference for large language models.

Extensibility : The system can be extended to support additional features such as:

- character generation
- story summaries
- dialogue generation

2.6 API Catalogue

API Endpoint

Generate Screenplay Scene

POST /generate

Request Format

```
{  "topic": "string",
 "genre": "string",
 "mood": "string",
 "scene_length": "string" }
```

Response Format

```
{ "topic": "string",
 "genre": "string",
 "mood": "string",
 "scene": "generated screenplay scene",
 "retrieved_context": "text" }
```

External AI API

Service: **Groq LLM API**

3. References

1. React Documentation
2. FastAPI Documentation
3. FAISS Documentation
4. Sentence Transformers Documentation
5. Groq API Documentation